THE BEGINNING

Module 10

Calibration & Tuning — Tier 2: Intermediate

1. Why Calibrate?

Every printer (even the same model) behaves slightly differently. Calibration tunes your specific machine and filament combination for more reliable, higher-quality prints.

2. Flow Rate Calibration

Ensures the right amount of plastic is extruded — too much causes blobbing/rough surfaces, too little causes weak, gappy prints. Most slicers have a built-in flow calibration test print.

3. Temperature Towers

A test print that changes nozzle temperature at each level, letting you visually compare quality (stringing, layer adhesion, surface finish) and pick the ideal temperature for a specific filament.

4. First-Layer Calibration

- Bed leveling — ensures consistent nozzle-to-bed distance across the whole surface
- Z-offset — fine-tunes exact starting height of the nozzle
- Most modern printers automate this, but manual fine-tuning still helps in tricky cases

5. When to Recalibrate

- After changing filament brands
- After any hardware changes (nozzle swap, bed swap)
- If print quality suddenly degrades unexpectedly

End of Module 10 — End of Tier 2. Next: Module 11, AI-Assisted Design